Evaluation of Remote Sensing Reflectance Derived From the Sentinel-2 Multispectral Instrument Observations Using POLYMER Atmospheric Correction

Minwei Zhang and Chuanmin Hu

***Abstract*— With a five-day revisit frequency over coastal regions and a spatial resolution of 10–60 m, the Sentinel-2 multispectral instrument (MSI) has shown its capacity to provide a reasonably accurate remote sensing reflectance (R_{rs}) data product over water when the standard "black pixel" (BP) atmospheric correction algorithm was applied to the top-of-atmospheric (TOA) reflectance data. Alternative atmospheric correction approaches, such as the POLYnomial-based algorithm applied to Medium Resolution Imaging Spectrometer (MERIS) (POLYMER), may show advantages under nonoptimal observation conditions (e.g., in the presence of strong sun glint). Here, POLYMER is implemented to process the data collected by both MSI and the Moderate Resolution Imaging Spectroradiometer (MODIS) with the resulting R_{rs} evaluated with concurrent and colocated *in situ* R_{rs} data collected from the AERONET-OC platforms. The results indicate less uncertainties in the MSI R_{rs} than those in the MODIS R_{rs}, and also less uncertainties in the MSI R_{rs} than those reported earlier. This is possibly attributed to the spatial heterogeneity of coastal waters where MODIS coarse-resolution data may suffer, and to the high-quality AERONET-OC data. In addition, for the evaluation data set, MSI R_{rs} does not appear to suffer from adjacency effects from the AERONET-OC platform and clouds, leading to more coverage than MODIS in nearshore waters. However, MSI R_{rs} is noisy in relatively clear waters, possibly due to the noisy TOA reflectance in the atmospheric correction bands over clear waters.**

***Index Terms*— AERONET-OC, atmospheric correction, Moderate Resolution Imaging Spectroradiometer (MODIS), multispectral instrument (MSI), ocean color, POLYnomial-based algorithm applied to Medium Resolution Imaging Spectrometer (MERIS) (POLYMER), Sentinel-2.**

I. Introduction

FOR satellite sensors designed specifically to measure the "color" (i.e., spectral reflectance) of the ocean [e.g., Sea-viewing Wide Field-of-view Sensor (SeaWiFS), Moderate Resolution Imaging Spectroradiometer (MODIS), Medium Resolution Imaging Spectrometer (MERIS)], the mission goal of obtaining surface remote sensing reflectance (R_{rs}) within 5% uncertainty for the blue bands over oligotrophic waters have been reached [1], [2], owing to system vicarious calibration, state-of-the-art atmospheric correction for clear waters, and relatively low aerosol loads. However, R_{rs} data product for coastal waters generally contains relatively high uncertainties because of the optical complexities in both atmosphere and water [3].

Ocean color data products typically have a spatial resolution of ∼1 km, making them difficult to use for nearshore waters or estuaries where optical properties may vary within meters or tens of meters [4]. This issue could partly be addressed through the use of Sentinel-2 multispectral instrument (MSI) observations, which provides five-day revisits at 10–60-m spatial resolution when combining Sentinel-2A and Sentinel-2B. MSI has shown its capacity of mapping near-surface water reflectance in coastal waters [5] when the traditional "black pixel" (BP) algorithm was used for atmospheric correction through two near-infrared (NIR) or shortwave infrared (SWIR) bands [6], [7].

NASA has been using the "BP" algorithm for the atmospheric correction of all mainstream ocean color sensors (therefore named as NASA standard atmospheric correction or NSAC). Iteration is used in NSAC to account for the signals in the two NIR bands from "non-black" waters [8]. While NSAC works well under optimal observing conditions [1], it may fail under nonoptimal conditions; for example, when strong sun glint is present [9]. In such cases, alternative atmospheric correction approaches may be used instead. One example is the POLYnomial-based algorithm applied to MERIS (POLYMER) [10], which is designed to process MERIS data even under the presence of strong sun glint. The concept is to apply polynomial functions to fit the atmospheric contribution instead of estimating atmospheric contribution through explicit aerosol modeling. POLYMER has been implemented to process the MODIS data for the coastal regions of the Gulf of Mexico, which showed two to three times more retrievals of R_{rs} with slightly higher uncertainties than NSAC [4]. Similarly, Steinmetz and Ramon [22] applied POLYMER to process MSI data, with the retrieved R_{rs} evaluated using *in situ* data collected by AERONET-OC platforms. The data are provided at three quality levels: Levels 1.0, 1.5, and 2.0. Level 1.0 means unscreened data, Level 1.5 means cloud cleared

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TABLE I
BAND SPECIFICATIONS OF MSI

	Sentinel-2A		Sentinel-2B		
Band	Central wavelength (nm)	Bandwidth (nm)	Central wavelength (nm)	Bandwidth (nm)	Spatial resolution (m)
1	442.7	21	442.2	21	60
2	492.4	66	492.1	66	10
3	559.8	36	559.0	36	10
4	664.6	31	664.9	31	10
5	704.1	15	703.8	16	20
6	740.5	15	739.1	15	20
7	782.8	20	779.7	20	20
8	832.8	106	832.9	106	10
8a	864.7	21	864.0	22	20

and quality-controlled data, and Level 2.0 means quality-assured data through additional quality assurance constraints. Currently, Level 2.0 data are used by SeaBASS for validating satellite data products. Because Level 1.5 data were used by Steinmetz and Ramon [22] and the AERONET-OC data used by SeaBASS for validation is Level 2.0, it is necessary to revisit the MSI R_{rs} data product uncertainties using Level 2.0 data. Furthermore, because MODIS R_{rs} data products from both the NSAC and POLYMER atmospheric corrections have been evaluated by the community [3], [4], [11], it is necessary to compare the uncertainties of MSI R_{rs} and MODIS R_{rs} using the same *in situ* data set. These represent the main objectives of this article.

Because of higher resolution (10–60 m) than MODIS (1-km), MSI can yield R_{rs} data closer to land than from MODIS, making it necessary to evaluate potential land adjacency effects on MSI R_{rs}. For MODIS, the pixels within a window of 7×5 (± 3 pixels along the scan, ± 2 pixels along the track) of bright targets (land or clouds) are flagged as being contaminated by straylight [12], which are discarded in the global composite data products [13], [14]. If the same constraint holds true for MSI, the advantage of using MSI to get closer to land will be significantly compromised. Therefore, the third objective of this article is to evaluate potential land or cloud adjacency effects on MSI R_{rs} retrieved using POLYMER. Finally, because MSI is known to have lower signal-to-noise ratios (SNRs) than MODIS, the capacity of MSI in deriving R_{rs} over dark waters is evaluated.

II. DATA AND METHODS

A. Data

L1C data from MSI of both Sentinel-2A and Sentinel-2B were downloaded from the United States Geological Survey (https://earthexplorer.usgs.gov/) and then processed using POLYMER, from which R_{rs} was derived. MSI bands used in this article are listed in Table I. MSI data were processed at the spatial resolution of 60 m unless otherwise specified. MSI bands with spatial resolutions of 10 and 20 m were binned into 60 m before applying the POLYMER atmospheric correction.

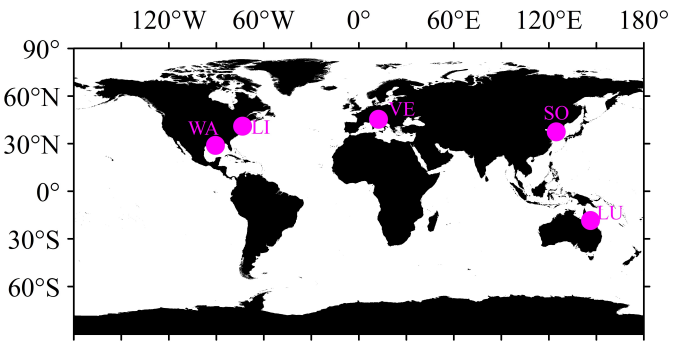

Fig. 1. AERONET-OC stations, where *in situ* R_{rs} data were collected and used in this article for evaluating MSI and MODIS R_{rs} retrievals. The letters represent first two names of the station, which are Venise (45.314°N, 12.508°E), LISCO (40.955°N, 73.342°W), Socheongcho (37.423°N, 124.738°E), WaveCIS_Site_CSI_6 (28.867°N, 90.483°W), and Lucinda (18.520°S,146.386°E).

Level-1B data from MODIS aboard satellite Aqua were obtained from the NASA Goddard Space Flight Center (https://oceancolor.gsfc.nasa.gov). POLYMER implemented for MODIS was used to retrieve R_{rs} from Level-1B data [4].

In situ R_{rs} collected over five AERONET-OC stations (Venise, LISCO, Socheongcho, WaveCIS_Site_CSI_6, and Lucinda, all annotated in Fig. 1) were used to evaluate MSI- and MODIS-retrieved R_{rs}. There are three quality levels of the *in situ* R_{rs} including Levels 1.0, 1.5, and 2.0. Readers are referred to Zibordi *et al.* [15] for a detailed description of the data levels. Level 2.0 data with the highest quality were downloaded from the AERONET-OC data archives (https://aeronet.gsfc.nasa.gov). R_{rs} in the wavelengths close to the central wavelengths of MSI or MODIS bands were used in the evaluation. Specifically, the AERONET-OC measurements in wavelengths 440/441/442 (wavelength measured at different AERONET-OC stations are slightly different), 501/491, 555/551, and 674/667/668 were used to evaluate MSI bands at 443, 492, 560, and 665 nm, respectively.

B. POLYMER Atmospheric Correction

Following the approach detailed by Steinmetz *et al.* (2011) [10], POLYMER was implemented in the atmospheric correction of MSI data, from which R_{rs} is retrieved. The bidirectional reflectance distribution function (BRDF) effect is accounted for based on the model presented by Morel and Gentili [16]–[18], where the Chl is estimated using R_{rs}. In order to do quality control of the retrieved R_{rs}, several criteria were used to exclude low-quality retrievals, including: 1) thick clouds, i.e., Rayleigh-corrected top-of-atmospheric (TOA) reflectance at 865 nm (denoted by $\rho_t'(865)$) > 0.05; 2) severe sun glint, i.e., sun glint coefficient $L_{gn} > 0.04$ sr^{-1} where L_{gn} was estimated for each pixel using National Centers

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for Environmental Prediction (NCEP) wind and the models described in [9] and [19]; and 3) Chl is out of bounds, i.e., <0.01 or >100 mg·m^{-3}.

C. Evaluation of MSI R_{rs} Using In Situ R_{rs}

While AERONET-OC R_{rs} covers wavelengths from blue to SWIR, only the short wavelengths are used due to the noise issue with the long wavelengths. Specifically, R_{rs} in wavelengths close to the MSI bands at 443, 492, 560, and 665 nm are utilized. MSI data from Sentinel-2A and Sentinel-2B were not separated in the evaluation. MSI R_{rs} were retrieved at spatial resolution of 60 m. A time window of 3 h between MSI overpass and *in situ* R_{rs} measurement was used to search for the matching pairs [20]. The MSI overpass times are around 10:10, 15:50, 02:30, 16:40, and 00:30 (UTC time) for Venise, LISCO, Socheongcho, WaveCIS_Site_CSI_6, and Lucinda, respectively. Following the method presented in [21], a uniformity screen is applied to calculate the mean retrieved R_{rs}. If the number of valid retrievals of MSI R_{rs} within a 3×3 box centered at the location of *in situ* measurement was greater than half of the total number of pixels in the box and the coefficient of variation (CV) among these pixels was $<15\%$, the mean R_{rs} from these valid retrievals was used to compare with the concurrent *in situ* R_{rs}. The same criteria were used to search for the match-up pixels in MODIS.

For the match-up comparison between MSI and *in situ* R_{rs}, four statistical measures were calculated to describe the difference. These include root mean square difference (RMSD), unbiased root mean square relative difference (uRMSRD), bias (δ), and spectral angle mean (SAM)

$$\mathrm{RMSD} = \sqrt{\frac{\sum_{i=1}^{n}\left(R^i_{\mathrm{rsm}} - R^i_{\mathrm{rsf}}\right)^2}{n}} \tag{1}$$

$$\mathrm{uRMSRD} = \sqrt{\frac{\sum_{i=1}^{n}\left(2 * \left(R^i_{\mathrm{rsm}} - R^i_{\mathrm{rsf}}\right)/\left(R^i_{\mathrm{rsm}} + R^i_{\mathrm{rsf}}\right)\right)^2}{n}} \tag{2}$$

$$\delta = \frac{\sum_{i=1}^{n}\left(R^i_{\mathrm{rsm}} - R^i_{\mathrm{rsf}}\right)}{n} \tag{3}$$

$$\mathrm{SAM} = \frac{\sum_{i=1}^{n}\arccos\left(\frac{\left\langle R^i_{\mathrm{rsm}}, R^i_{\mathrm{rsf}}\right\rangle}{\left\|R^i_{\mathrm{rsm}}\right\| * \left\|R^i_{\mathrm{rsm}}\right\|}\right)}{n} \tag{4}$$

where R_{rsm} and R_{rsf} represent R_{rs} from MSI and field measurement, respectively, and i refers to ith matching pair. n is the total matching pairs. $\langle R^i_{\mathrm{rsm}}, R^i_{\mathrm{rsf}}\rangle$ is the dot product of the MSI and field measured R_{rs} spectra. $\|R_{\mathrm{rsm}}\|$ and $\|R_{\mathrm{rsf}}\|$ are the Euclidean norms of MSI-retrieved and field-measured R_{rs} spectra. The uRMSRD was used to account for the potential errors in R_{rsf} when R_{rsf} is small; otherwise, a small error in R_{rsm} could lead to an unrealistically high relative difference.

D. Adjacency Effects on R_{rs}

Ideally, land adjacency effects on MSI R_{rs} should be evaluated from waters close to land. However, it is impossible to evaluate whether the changes in R_{rs} (as a function of distance from land) are due to land adjacency effects or due to realistic gradients in R_{rs}, as waters closer to shoreline typically have higher concentrations of water constituents (phytoplankton, suspended nonliving particles, and/or colored dissolved organic matter). Therefore, the AERONET-OC observation platform "WA" shown in Fig. 1 was used as a surrogate for land. This is because the AERONET-OC station is in the offshore region (https://aeronet.gsfc.nasa.gov/new_web/ocean_levels_versions.html), where optical properties of the water around the station are not expected to vary a lot within a short distance. Feng and Hu [14] indicated a decreasing pattern of land adjacency effects on the SWIR top of atmosphere radiance by increasing the distance to land; however, the decreasing pattern appears to be universal in pixel space (rather than in physical distance space) among the sensors evaluated (including MODIS, TM, ETM+, and OLI). This is certainly against intuition but is based on the statistics of a large quantity of data. Feng and Hu [14] also indicated that land adjacency effects on the SWIR TOA radiance are negligible for waters >12 pixels distance from land. In this article, we chose a pixel that is 15 pixels distance from the AERONET-OC observation platform. This pixel could be reasonably assumed free of platform adjacency effects on both MSI TOA radiance and MSI R_{rs}. Note that in order to evaluate the adjacency effects, the optical properties of water should not be highly variable within these 15 pixels. As a result, we did not choose a pixel too far away from the platform, since the farther the pixel, the higher the variability of optical properties. Fig. 2 shows the specification of the pixels used to evaluate potential adjacency effects. The R_{rs} ratio between the current pixel and pixels outside the ± 15 pixel boundary was used to evaluate the potential adjacency effect, with a ratio of 1.0 denoting free of adjacency effects. The cloud adjacency effects were evaluated using the same way, where cloud pixels in the offshore regions were identified using a threshold for $\rho'_t(865) > 0.05$.

Fig. 2. Transect line showing the land/cloud pixels as well as the adjacent ocean pixels.

III. Results

A. Evaluation of MSI R_{rs}

MSI R_{rs} retrieved using POLYMER is first evaluated by comparing it with concurrent *in situ* R_{rs} from AERONET-OC measurements and then by comparing it with MODIS R_{rs} retrieved using POLYMER. Fig. 3 shows the first comparison results, and the statistical measures are listed in Table II. A total of 28 MSI and 25 MODIS scenes were used for the evaluation. Fig. 3 shows that one satellite-retrieved R_{rs} may

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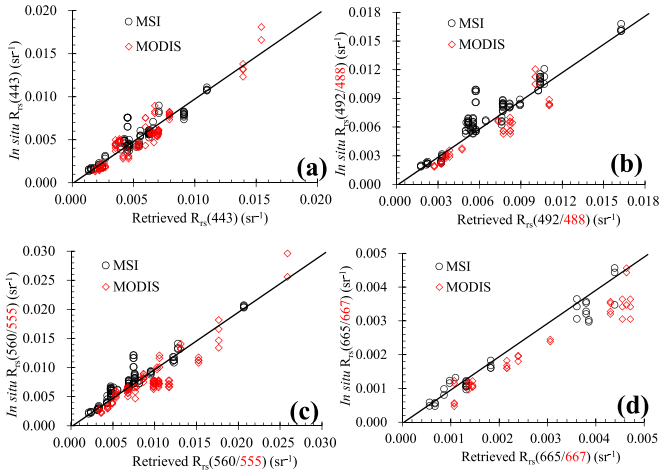

Fig. 3. Scatter plots of MSI and MODIS retrieved R_{rs} versus AERONET-OC *in situ* R_{rs} for bands at (a) 443 nm, (b) 492 nm for MSI and 488 nm for MODIS, (c) 560 nm for MSI and 555 nm for MODIS, and (d) 665 nm for MSI and 667 nm for MODIS. It is noted that one satellite-retrieved R_{rs} may correspond to several *in situ* R_{rs} points because of multiple *in situ* measurements within a time window of ±3 h. Statistical measures of uncertainties in the MSI and MODIS R_{rs} are listed in Table II.

TABLE II

STATISTICAL MEASURES OF UNCERTAINTIES IN THE MSI AND MODIS R_{rs}. THE GRAPHICAL FORMS FOR MSI ARE SHOWN IN FIGS. 3 AND 4

Band		N[a]		RMSD (10^{-4} sr^{-1})		uRMSRD (%)		δ (10^{-4} sr^{-1})	
MSI	MOD[b]	MSI	MOD	MSI	MOD	MSI	MOD	MSI	MOD
443	443	97	96	9.4	10.2	17.3	22.1	0.83	3.2
492	488	97	36	11.5	16.5	17.2	28.0	-4.5	12.4
560	555	94	96	12.3	22.4	15.9	26.4	-4.0	15.3
665	667	34	35	3.6	7.4	17.5	31.7	2.1	6.1

[a] N is the No. of matching pairs.

[b] MOD means MODIS.

correspond to several *in situ* R_{rs} points due to multiple *in situ* measurements within a time window of ±3 h. The multiple *in situ* measurements were not reduced to a single matchup in order to increase the number of matchups, making the evaluation results more statistically significant. Although MODIS R_{rs} retrieved from POLYMER have been evaluated using *in situ* data over North American waters in [4], for apples-to-apples comparison they are further evaluated using the same *in situ* data set here, with results also shown in Fig. 3 and statistical measures also listed in Table II. The SAM for the matchup comparison between MSI retrieved and *in situ* R_{rs} is 3.2°. The number is 5.1° for MODIS-retrieved R_{rs} versus *in situ* R_{rs}. The smaller numbers of matching pairs for MSI band 665 nm and MODIS bands 488- and 667-nm result from lack of *in situ* R_{rs} measured in the wavelengths close to these bands in the AERONET-OC station Venise, i.e., the *in situ* R_{rs} wavelength at 674 nm is far from the central wavelength of the MSI band 665 nm (bandwidth of 33 nm) and the MODIS band 667 nm (bandwidth of 10 nm). The AERONET-OC *in*

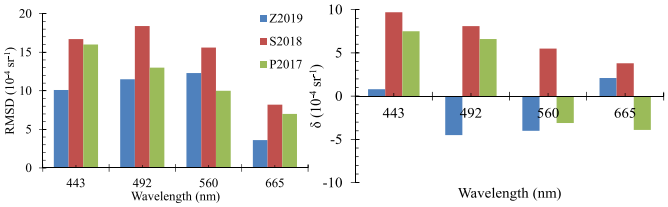

Fig. 4. RMSD and δ of MSI R_{rs} retrieved from the POLYMER approach presented in this article (denoted by Z2019), compared with those also from the POLYMER approach earlier by Steinmetz and Ramon [22] (denoted by S2018) and from the "BP" approach by Pahlevan *et al.* [5] (denoted by P2017).

situ R_{rs} wavelength of 501 nm is far from the MODIS band 488 nm (bandwidth of 10 nm). AERONET-OC *in situ* R_{rs} data at 501 nm is used to evaluate the MSI R_{rs} data at 492 nm (bandwidth of 66 nm). Based on the results shown in Fig. 3 and Table II, MSI R_{rs} appears to have a slightly lower uncertainty than MODIS R_{rs}.

Fig. 4 shows the graphical form of the statistical measures of uncertainties in MSI R_{rs}. In comparison, uncertainties estimated earlier by Steinmetz and Ramon [22] and by Pahlevan *et al.* [5] are also shown in Fig. 4. In these earlier evaluations, POLYMER was used by Steinmetz and Ramon [22] for retrieving R_{rs} from the MSI data, whereas the "BP" atmospheric correction (using the 865- and 1613-nm bands) was used by Pahlevan *et al.* [5] for the same purpose. Although POLYMER was used both in this article and in [22], the statistical measures are different from lower uncertainties in MSI R_{rs} found in this article than in [22]. The different statistical measures are possible because of the following.

1) The different AERONET-OC *in situ* R_{rs} data used in the evaluations. The level of AERONET-OC data is different, i.e., Level 2.0 in this article and Level 1.5 for Steinmetz and Ramon [22]. The AERONET-OC sites are different.
2) Different sizes of the box were used to calculate the average of MSI R_{rs}, which is compared with *in situ* data. A box of 3 × 3 was used in this article and 5 × 5 by Steinmetz and Ramon [22]. In addition to the improved performance when using the same POLYMER atmospheric correction, Fig. 4 also suggests slightly lower uncertainties in the current MSI R_{rs} than those retrieved from the "BP" atmospheric correction.

Fig. 5 shows a visual comparison between MSI R_{rs} and MODIS R_{rs}, both retrieved through POLYMER. The same 3 km × 3 km window was used to mask stray light from the land. The spatial variation patterns agree very well between MSI and MODIS R_{rs}. As expected, MSI R_{rs} shows more details than MODIS R_{rs} due to their resolution difference (∼60 m versus ∼1000 m), as shown in Fig. 6. The frequency distributions of MSI R_{rs} and MODIS R_{rs} show the slight difference (see Fig. 7), possibly due to the different band specifications (e.g., bandwidth, central wavelength) and the temporal variation of water's optical properties between MSI and MODIS overpasses (15:53 and 18:10 GMT, respectively). Despite such a difference, Figs. 5–7 suggest relatively con-

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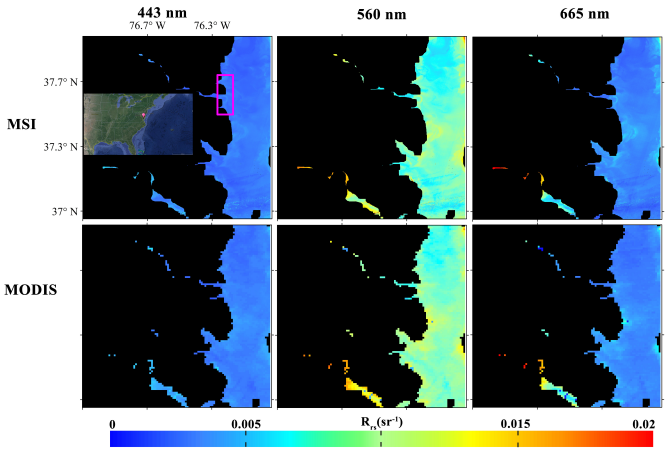

Fig. 5. R_{rs} retrieved from the MSI data and the MODIS data on October 23, 2018, using POLYMER. The wavelengths annotated on the top of the panel indicate MSI bands with the corresponding MODIS bands at 443, 555, and 667 nm, respectively. The overpass times for MSI and MODIS are 15:53 and 18:10 GMT, respectively. A small region outlined by the rectangle is shown in Fig. 6.

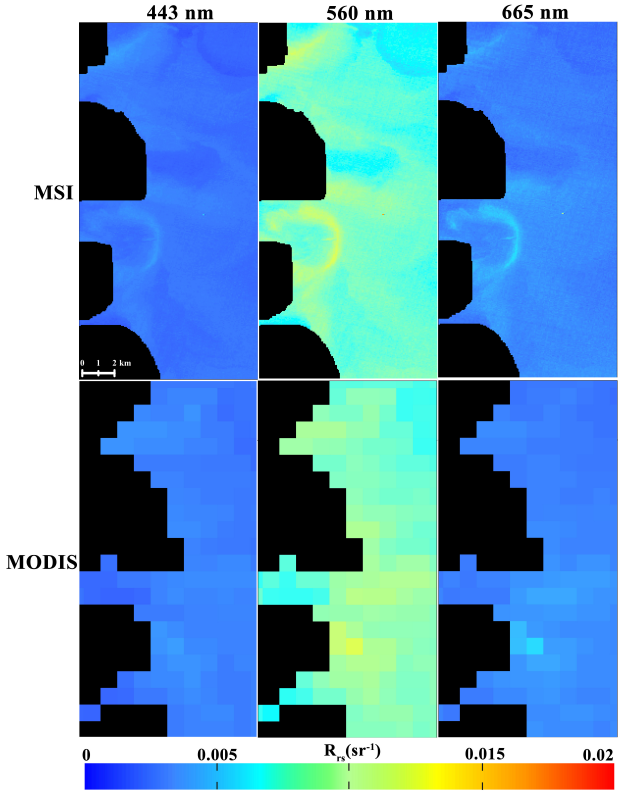

Fig. 6. Same as Fig. 5, but the small region outlined by the rectangle of Fig. 5 is enlarged to show the details here.

sistent results between MODIS and MSI R_{rs} retrieved using POLYMER.

While the same mask was used in Fig. 5 to compare MSI and MODIS R_{rs}, Fig. 8 shows the R_{rs} retrievals without any mask. Here, water pixels were identified using the Normalized Difference Water Index (NDWI) [23]. Clearly, MSI R_{rs} shows great details near land due to its high spatial resolution (see

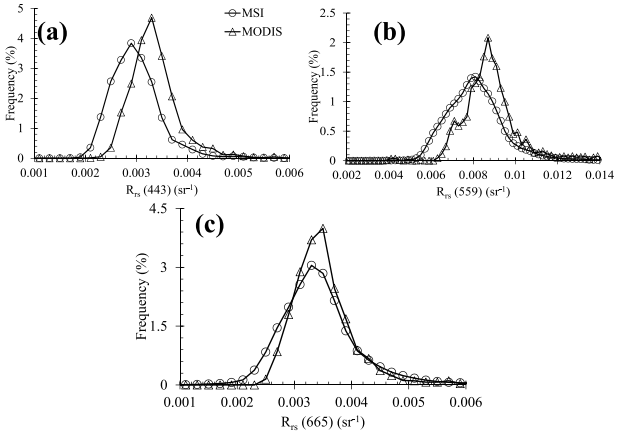

Fig. 7. Frequency distribution of R_{rs} over the region in Fig. 5. (a) MSI versus MODIS 443 nm, (b) MSI 560 nm versus MODIS 555 nm, and (c) MSI 665 nm versus MODIS 667 nm.

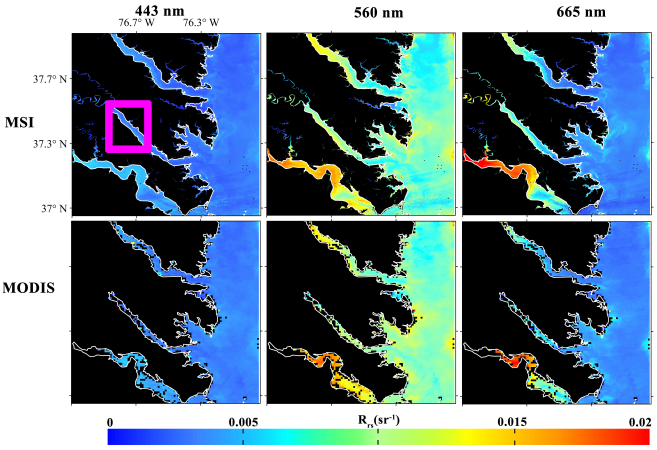

Fig. 8. Same as Fig. 5 but without masking pixels near land. A small region outlined by the rectangle is shown in Fig. 9. White line is the coastal line.

Figs. 8 and 9). On the contrary, MODIS pixels near land could be mixed pixels, leading to either no retrievals or erroneous retrievals.

B. Adjacency Effects on MSI Retrieved R_{rs}

Fig. 10(a) and (b) shows the ratio of R_{rs} over pixels numbered 1–15 (−1 to −15) to R_{rs} over a pixel numbered 15 (−15) derived from six MSI images in four seasons around the "WA" station annotated in Fig. 1. Overall, there are slight pixel-to-pixel variations but there is a lack of overall trend away from the platform. This is different from the land adjacency effects observed from MODIS TOA observations [14], which show a decreasing trend away from the land. Therefore, Fig. 10(a) and (b) indicates that the effects of the AERONET-OC platform on MSI R_{rs} retrieved from POLYMER are negligible.

However, this does not mean that land adjacency effects on MSI R_{rs} are negligible. This is because land may be much larger than the AERONET-OC platform. Because it is extremely difficult to find land with homogeneous adjacent water pixels to evaluate the possible land-adjacency effects, clouds are used as surrogates of land to evaluate cloud-

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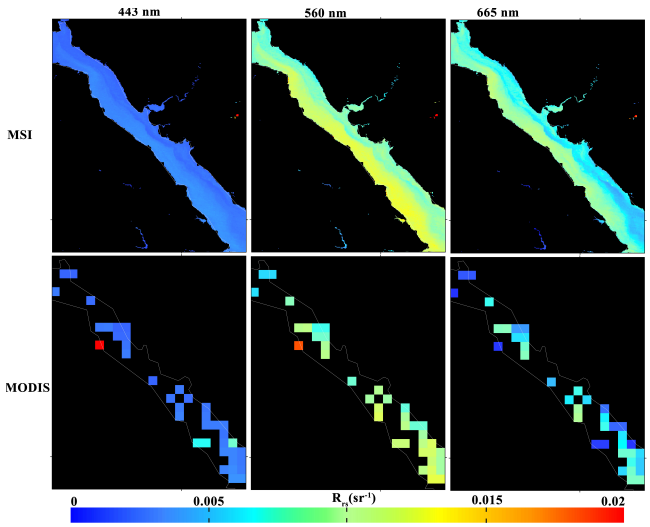

Fig. 9. Same as Fig. 8 but over the small region outlined in Fig. 8, which is enlarged to show the details. White line is the coastal line.

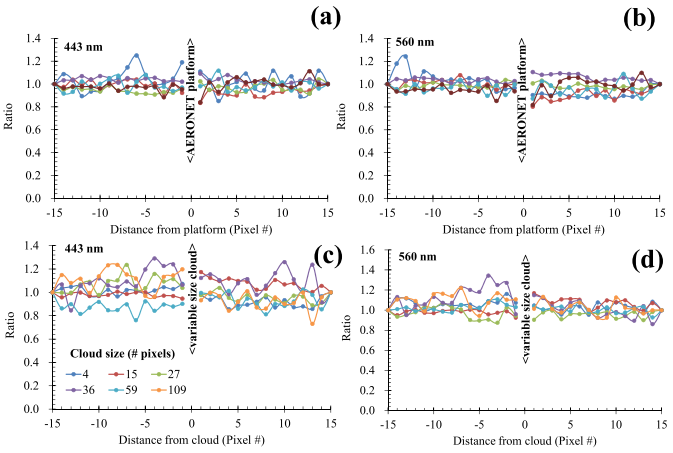

Fig. 10. MSI R_{rs} ratio between platform-adjacent pixels and platform-away pixels for (a) 443 and (b) 560 nm. (c) and (d) Same around a cloud patch. The ratio is defined as between the current pixel and pixels away from the platform (or cloud) by >15 pixels. The pixel # is illustrated in Fig. 2. Six MSI images in four seasons were used in (a) and (b), while six cloud patches with the size from a couple of pixels to hundreds of pixels were used in (c) and (d).

adjacency effects. Using a similar way as shown in Fig. 10(a) and (b), the cloud adjacency effects are evaluated by calculating the ratio of R_{rs} over cloud-adjacent pixels (1–15 pixel away from cloud edge) to R_{rs} over the pixel that is 15 pixels away from the cloud edge, with the results shown in Fig. 10(c) and (d), respectively. The six examples represent different sizes of clouds, ranging from a few pixels to hundreds of pixels. Similar to the argument for the AERONET-OC platform mentioned above, because there is no systematic and consistent R_{rs} trend away from clouds, one may infer that cloud-adjacency effects are not apparent, possibly because the adjacency effects on the TOA radiance are mostly compensated through the POLYMER atmospheric correction.

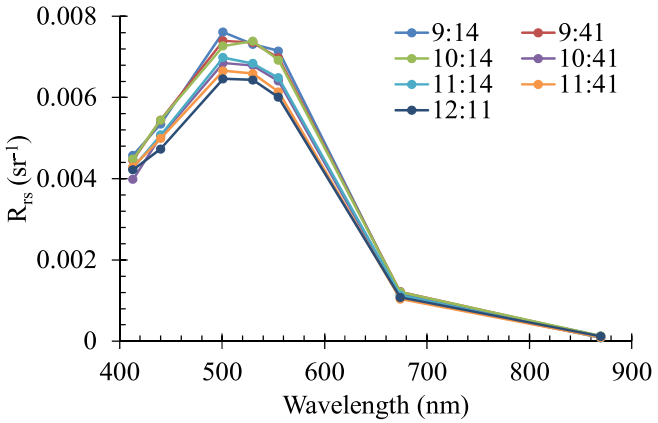

Fig. 11. R_{rs} measured at the station Venise during 3 h on May 16, 2017. The time in the legend is UTC time.

IV. Discussion

A. Uncertainties Sources

Similar to other validation studies, the differences between satellite-derived and *in situ*-measured R_{rs} can come from multiple sources. Uncertainties in the satellite-derived R_{rs} may be a result of imperfect radiometric calibration, atmospheric correction, and sensor noise. Uncertainties in the *in situ* R_{rs} may be a result of imperfect instrument calibration, difficulties in accurately removing surface-reflected skylight (Fresnel reflection) [24], among other uncertainties in data reduction procedures. Such uncertainties may be in the range of 3%–5% even under ideal situations [20], [25]. Therefore, the RMSD and uRMSRD values between satellites and AERONET-OC data not only originated from imperfect satellite measurements/retrievals; however, a small portion resulted from *in situ* measurements and data reductions.

Temporal variations of water's optical properties may also contribute to the RMSD and uRMSRD values mentioned above, due to the different times between satellite and *in situ* measurements. Because of the 3-h time window used, one MSI measurement may correspond to several *in situ* measurements (see Fig. 3), whose variations may not be negligible. Fig. 11 shows such an effect where the relative variation of R_{rs} at AERONET-OC station Venise during 3 h on May 16, 2017, reached 15%.

Spatial variations of water's optical properties may also contribute to the RMSD and uRMSRD values mentioned above. Although there were no flow-through measurements to quantify spatial variability of water properties near the AERONET-OC stations, it is expected that spatial variability in coastal waters may not be negligible even at the subpixel scale. The spatial homogeneity test (i.e., CV < 15% from the 3 × 3 pixel box) only assures cross-pixel homogeneity, where subpixel variability cannot be ruled out by the CV criteria. This is perhaps the reason why MSI R_{rs} showed slightly lower uncertainties than MODIS R_{rs} (see Table II). In any case, these uncertainties are generally comparable to the MODIS performance over coastal waters [3] and, in particular, uncertainties in the MSI R_{rs} data are lower than those reported

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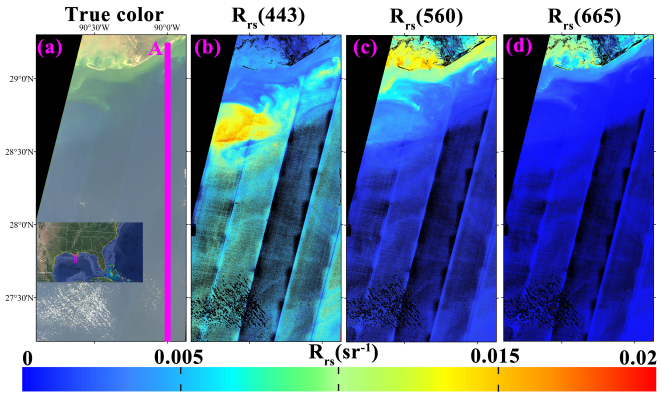

Fig. 12. (a) True-color RGB composite of the MSI data on February 22, 2018 (R: 665 nm, G: 560 nm, B: 443 nm) and the corresponding R_{rs} retrievals from POLYMER at (b) 443, (c) 560, and (d) 665 nm. The solid line in (a) is used to show variations of CV in both ρ_t and R_{rs} along this line. The pepper noise (black dots) in offshore waters represents negative R_{rs} values.

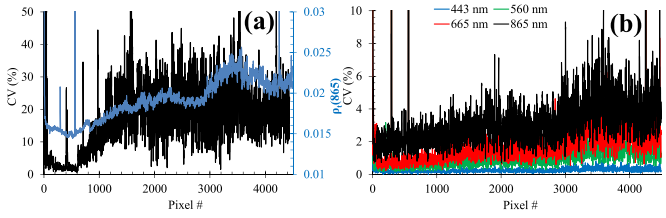

Fig. 13. (a) CV of $R_{rs}(443)$ and mean $\rho_t(865)$. (b) CV of ρ_t over 3 × 3 pixels along a transect line starting from pixel "A" in Fig. 12(a).

earlier [5], [22], suggesting that the MSI R_{rs} data are suitable for coastal water applications.

B. Advantages and Disadvantages of MSI R_{rs}

In addition to the slightly better performance over MODIS, MSI has the advantage of revealing more details of water patterns even near shorelines due to its significantly improved spatial resolution (see Figs. 6 and 9). This is particularly important for small estuaries where MODIS often yields no retrievals. The lack of apparent adjacency effects in POLYMER-retrieved R_{rs} makes the MSI data even more valuable for estuarine and coastal waters.

However, two major disadvantages do exist with the MSI data. First, for the same reason for revealing more details in estuarine and coastal waters, the high spatial resolution is accompanied by low revisit frequency. Even with the combined Sentinel-2A and Sentinel-2B, the revisit frequency of once every five days is still much lower than MODIS. For fast-changing coastal waters, this revisit frequency may not be enough to derive long-term mean and trend or may miss some algae blooms of ephemeral nature. Second, although not designed for open oceans, MSI does have coverage over some offshore waters. However, the MSI R_{rs} data are very noisy (see Fig. 12) over such waters. For example, the noise in $R_{rs}(443)$ can be quantified by the CV [see Fig. 13(a)] over a running window of 3 × 3 pixels along the transect annotated by a solid line in Fig. 12. In addition to apparent vertical banding, invalid R_{rs} retrievals frequently occurred in offshore clear waters, which are indicated by negative R_{rs} values (black dots). This is possibly a result of noisy ρ_t at 865 nm, i.e., the CV of $\rho_t(865)$ can reach 6% [see Fig. 13(b)]. Because $R_{rs}(865)$ from offshore clear waters is essentially zero, $\rho_t(865)$ is typically lower than over more turbid coastal waters, leading to higher noise. Higher noise in the NIR atmospheric correction bands would lead to higher uncertainties in the retrieved R_{rs} [26], [27]. An attempt was made to find a threshold in $\rho_t(865)$ to partition the MSI R_{rs} data into "trustable" and "non-trustable" but the attempt did not lead to meaningful results because $\rho_t(865)$ also changes with solar/viewing geometry, aerosols, and sun glint. Future efforts should be dedicated to defining the conditions leading to successful and nonsuccessful R_{rs} retrievals. At the present time, the MSI R_{rs} data over offshore clear waters should be used with great caution.

V. Conclusion

A spectral-matching atmospheric correction approach (POLYMER) was implemented to process the MSI data, where the retrieved R_{rs} data were evaluated against *in situ* R_{rs} data collected at AERONET-OC stations and against MODIS R_{rs} data retrieved also through POLYMER. MODIS-retrieved R_{rs} were also evaluated by the same *in situ* data set. Results indicate lower uncertainties in MSI R_{rs} than in MODIS R_{rs}(e.g., RMSDs are 9.4, 11.5, 12.3, and 3.6 sr^{-1} for MSI bands at 443, 492, 560, and 665 nm, lower than that for MODIS bands, which are 10.2, 16.5, 22.4, and 7.4 sr^{-1}), and also lower uncertainties in MSI R_{rs} than from earlier evaluations where either POLYMER or standard NASA atmospheric correction approaches were applied. The lower uncertainties than previous evaluations for POLYMER presented by Steinmetz and Ramon [22] is possibly due to the high-level *in situ* data (means high accuracy) used in this study. Adjacency effects from bright targets on MSI R_{rs} retrieved from POLYMER are not obvious, indicating high-fidelity MSI data over coastal waters where detailed spatial patterns, even close to land, can be revealed. However, data from offshore clear waters need to be used with great caution due to significant banding and other noise.

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